Introduction to Open-Source Software

B. Tech Integrated /Computer / Sem IV

PART A

(PART A: TO BE REFFERED BY STUDENTS)

Experiment No.08

A.1—Aim:

Introduction to Libre Office tools.

A.2--- Prerequisite:

LibreOffice - formerly known as OpenOffice.org – is a freely available suite of office applications that you can download and install for free without any copyright infringement.OpenOffice.org has been taken over by Oracle and is no longer open-source software. LibreOffice suite was developed in 2010 by The Document Foundation – a non-profit organisation that promotes open-source document handling software. LibreOffice suite is a fork of OpenOffice, meaning the underlying source code is the same, but the software has gone in a different development direction. LibreOffice is a comprehensive, professional-quality office tool available in more than 114 languages and for all major operating systems, including Microsoft Windows, FreeBSD, Mac OSX and Linux (Debian, Ubuntu, Fedora, Mandriva, Suse, ...). LibreOffice is user-friendly. It offers a simple-to-use yet powerful interface that is easy to personalise. Its native file format is Open Document Format (ODF), an open standard format. Since it stores data in an international open standard format, it is therefore compatible with all major file formats.

LibreOffice Components

LibreOffice suite components include word process, spreadsheet, presentation graphics and database, drawing and math tools.

Office Tools

- Writer: It is a word processor and a text editor, similar to Microsoft Word or WordPerfect. Usually a file created by Writer is saved as ODT (Open Document File Text Document) with an .ODT file format. Writer can also read and edit files from other word processors with .DOC, .DOCX, .RTF, etc. file formats.
- Calc: It is a spreadsheet application, similar to Microsoft Excel or Lotus 1-2-3. It is a tool for creating and editing data sheets. Based on data series, it can also

generate different charts. Usually, a file created by Calc is saved as ODS (Open Document File Spreadsheet Document) with an .ODS file format. Calc can also read and edit files from other spreadsheet applications with .XLS, .XLSX, .XLT, etc. file formats.

- Impress: It is a presentation application, similar to Microsoft PowerPoint or Apple Keynote. Usually, a file created by Impress is saved as ODP (Open Document File Presentation Document) with an .ODP file format. Impress can also read and edit files from other presentation applications with .PPT, .PPTX, .PPS, etc. file formats.
- Base: It is a database management application, similar to Microsoft Access or FoxPro.
- Draw: It is a drawing editor, similar to Microsoft Paint or CorelDraw.
- Math. It is a tool for creating and editing mathematical formulae

A.3--- Tasks:

1. An article, regarding the impacts of climate change is given in the same folder in with the file name, Climate_Change.ott. Open this file in LibreOffice Writer and create a new **heading style** with the name **style21** defining the following features. (The new style has to be created under **Heading 2** in the **Heading styles**)

Font family - Bitstream Charter
Font Size - 105 % or 16pt
Font color - Purple
Add Shadow effect.

Add the style, **style21** to all the subheadings in the article. (Subheadings are marked in green colour)
Save the modified file.

2. An article, describing the major freedom struggles of Indian Independence Movement is given in the same folder with the file name, Freedom_Struggle.ott. Open this file in LibreOffice Writer and add table of contents in page 2. The table of contents has to be inserted using the **Table of Contents and Index** option of LibreOffice Writer.
Hints:
• Use the style **Heading 1** to subheadings. (Subheadings are marked in blue colour)
Save the modified file.

(PART - B)

(TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the Blackboard or emailed to the concerned lab in charge faculties at the end of the practical in case there is no Black board access available)

Roll.No. : C119	Name: Harshi Maniar
Sem/Year : IV / 2nd	Batch: C2
Date of Experiment :	Date of Submission:
Grade --	

B.1: Procedure of performed experiment

Effects of climate change

The effects of climate change span the physical environment, ecosystems and human societies. It also includes the economic and social changes which stem from living in a warmer world. Human-caused climate change is one of the threats to sustainability.

Many physical impacts of climate change are already visible, including extreme weather events, glacier retreat, changes in the timing of seasonal events (e.g., earlier flowering of plants), sea level rise, and declines in Arctic sea ice extent. The ocean has taken up between 20–30% of human-induced atmospheric carbon dioxide since the 1980s, leading to ocean acidification. The ocean is also warming and since 1970 has absorbed more than 90% of the excess heat in the climate system.

Physical impacts

A broad range of evidence shows that the climate system has warmed. Increasing temperature over land and the ocean, and sea level rise. The decreased snow cover in the Northern Hemisphere, and declining Arctic sea ice, both of which are indicative of global warming. Evidence of warming is also apparent in living (biological) systems such as changes in distribution of flora and fauna towards the poles.

Human-induced warming could lead to large-scale, abrupt and/or irreversible changes in physical systems. An example of this is the melting of ice sheets, which contributes to sea level rise and will continue for thousands of years. The probability of warming having unforeseen consequences increases with the rate, magnitude, and duration of climate change.

Wildlife and nature

Recent warming has strongly affected natural biological systems. Species worldwide are moving poleward to colder areas. On land, species move to higher elevations, whereas marine species find colder water at greater depths. Of the drivers with the biggest global impact on nature, climate change ranks third over the five decades before 2020, with only change in land use and sea use, and direct exploitation of organisms having a greater impact.

Regional effects

Regional effects of global warming vary in nature. Some are the result of a generalised global change, such as rising temperature, resulting in local effects, such as melting ice. In other cases, a change may be related to a change in a particular ocean current or weather system. In such cases, the regional effect may be disproportionate and will not necessarily follow the global trend.

There are three major ways in which global warming will make changes to regional climate: melting or forming ice, changing the hydrological cycle (of evaporation and precipitation) and changing currents in the oceans and air flows in the atmosphere. The coast can also be considered a region, and will suffer severe impacts from sea level rise.

On humans

The effects of climate change, in combination with the sustained increases in greenhouse gas emissions, have led scientists to characterize it as a climate emergency.It is an existential threat to civilization. Some areas may become too hot for humans to live in while people in some areas may experience displacement triggered by flooding and other climate change related disasters. The quality and quantity of freshwater will likely be affected almost everywhere. Some people may be particularly at risk from climate change, such as the poor, young children and the elderly. According to the World Health Organization, between 2030 and 2050, "climate change is expected to cause about 250,000 additional deaths per year." As global temperatures increase, so does the number of heat stress, heatstroke, and cardiovascular and kidney disease deaths and illnesses. When air pollution worsens, so does respiratory health, particularly for the 300 million people worldwide living with asthma; there is more airborne pollen and mold to torment hay fever and allergy sufferers.

INDIA'S STRUGGLE FOR INDEPENDENCE
1957-1947

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India's Struggle for Independence

India became independent from the British Rule on the 15th August 1947. While attaining that goal, India had witnessed many struggles leaded by our great freedom fighters. Here we are discussing some of the major events/struggles against the British Rule.

1957 – India's First Revolt of Independence

It is also termed Sepoy Riots by the British was an attempt to unite India against the invading British.

Pic 1: Rani Laxmibai

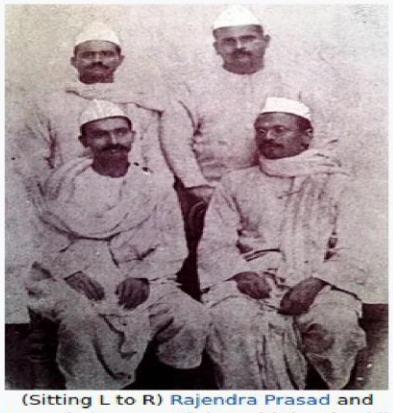
The Indian Rebellion of 1857 was a major, but ultimately unsuccessful, uprising in India in 1857–58 against the rule of the British East India Company, which functioned as a sovereign power on behalf of the British Crown. The rebellion began on 10 May 1857 in the form of a mutiny of sepoys of the Company's army in the garrison town of Meerut, 40 mi (64 km) northeast of Delhi (now Old Delhi). It then erupted into other mutinies and civilian rebellions chiefly in the upper Gangetic plain and central India, though incidents of revolt also occurred farther north and east. The rebellion posed a considerable threat to British power in that region, and was contained only with the rebels' defeat in Gwalior on 20 June 1858. On 1 November 1858, the British granted amnesty to all rebels not involved in murder, though they did not declare the hostilities to have formally ended until 8 July 1859. Its name is contested, and it is variously described as the Sepoy Mutiny, the Indian Mutiny, the Great Rebellion, the Revolt of 1857, the Indian Insurrection, and the First War of Independence

The important leaders who took part in the revolt were Mangal Pandey (Barrackpore), Bahadur Shah II (Delhi), Begum Hazrat Mahal (Lucknow), Rani Laxmibai (Jhansi).

1917 – Champaran Satyagraha

The Champaran Satyagraha of 1917 was the first Satyagraha movement led by Gandhi in India and is considered a historically important revolt in the Indian Independence Movement. It was a farmer's uprising that took place in Champaran district of Bihar, India, during the British colonial period. The farmers were protesting against having to grow indigo with barely any payment for it.

When Gandhi returned to India from South Africa in 1915, and saw peasants in northern India oppressed by indigo planters, he tried to use the same methods that he had used in South Africa to organize mass uprisings by people to protest against injustice.



(Sitting L to R) Rajendra Prasad and Anugrah Narayan Sinha, with local vakils Ramnavmi Prasad and Shambhusaran Varma (Standing L to R) during Mahatma Gandhi's 1917 Champaran movement.

Champaran Satyagraha was the first popular satyagraha movement. The Champaran Satyagraha gave direction to India's youth and freedom struggle, which was tottering between moderates who prescribed Indian participation within the British colonial system, and the extremists from Bengal who advocated the use of violent methods to topple the British colonialists in India.

Under Colonial-era laws, many tenant farmers were forced to grow some indigo on a portion of their land as a condition of their tenancy. This indigo was used to make dye. The Germans had invented a cheaper artificial dye so the demand for indigo fell. Some tenants paid more rent in return for being let off having to grow indigo. However, during the First World War the German dye ceased to be available and so indigo became profitable again. Thus many tenants were once again forced to grow it on a portion of their land- as was required by their lease. Naturally, this created much anger and resentment.

1920 – Non Corporation Movement

The Non-cooperation movement was launched on 5th September, 1920 by Mahatma Gandhi with the aim of self-governance and obtaining full independence (Purna Swaraj) as the Indian National Congress (INC) withdrew its support for British reforms following the Rowlatt Act of 21 March 1919, and the Jallianwala Bagh massacre of 13 April 1919.

The Rowlatt Act of March 1919, which suspended the rights of political prisoners in sedition trials,[1] was seen as a "political awakening" by Indians and as a "threat" by the British. Although it was never invoked and declared void just a few years later, the act motivated Gandhi to conceive the idea of satyagraha (truth), which he saw as synonymous with independence. This idea was also authorised the following month by Jawaharlal Nehru, for who the massacre also endorsed “the conviction that nothing short of independence was acceptable”.

Gandhi's planning of the non-cooperation movement included persuading all Indians to withdraw their labour from any activity that "sustained the British government and economy in India", including British industries and educational institutions. In addition to promoting “self-reliance” by spinning khadi, buying Indian-made goods only and boycotting British goods, Gandhi's non-cooperation movement called for the restoration of

the Khilafat (Khilafat movement) in Turkey and the end to untouchability. This result in public held meetings and strikes (hartals) led to the first arrests of both Jawaharlal Nehru and his father, Motilal Nehru, on 6 December 1921.

1929 – Purna Swaraj Declaration

The Purna Swaraj declaration (Hindi: पूर्ण, Purna, meaning "complete", Hindi: स्व, Swa, meaning "self" and Hindi: राज, raj, meaning "rule"), or Declaration of the Independence of India, was promulgated by the Indian National Congress on 26 January 1930, resolving the Congress and Indian nationalists to fight for Purna Swaraj, or complete self-rule independent of the British Empire.

Pic 2: The flag adopted in 1931

The flag of India was hoisted by Jawaharlal Nehru on 31 December 1929 on the banks of Ravi river, in Lahore, modern-day Pakistan. The Congress asked the people of India to observe 26th of January as Independence Day (see Legacy). The flag of India was hoisted publicly across India by Congress volunteers, nationalists and the public.

1930 – Civil Disobedience Movement

Britain's Salt Acts prohibited Indians from collecting or selling salt, a staple in the Indian diet. Citizens were forced to buy the vital mineral from the British, who, in addition to exercising a monopoly over the manufacture and sale of salt, also exerted a heavy salt tax. Although India's poor suffered most under the tax, Indians required salt. Defying the Salt Acts, Gandhi reasoned, would be an ingeniously simple way for many Indians to break a British law non-violently. He declared resistance to British salt policies to be the unifying theme for his new campaign of satyagraha, or mass civil disobedience.

Pic 3: Salt March

The Salt March, also known as the Salt Satyagraha, Dandi March and the Dandi Satyagraha, was an act of nonviolent civil disobedience in colonial India led by Mahatma Gandhi. The 24-day march lasted from 12 March 1930 to 5 April 1930 as a direct action campaign of tax resistance and nonviolent protest against the British salt monopoly. Another reason for this march was that the Civil Disobedience Movement needed a strong inauguration that would inspire more people to follow Gandhi's example. Mahatma

Gandhi started this march with 78 of his trusted volunteers. Walking about 18 km a day, the march spanned 390 km, from Sabarmati Ashram to Dandi, which was called Navsari at that time (now in the state of Gujarat). Growing numbers of Indians joined them along the way. When Gandhi broke the British Raj salt laws at 6:30 am on 6 April 1930, it sparked large scale acts of civil disobedience against the salt laws by millions of Indians.

After making the salt by evaporation at Dandi, Gandhi continued southward along the coast, making salt and addressing meetings on the way. The Congress Party planned to stage a satyagraha at the Dharasana Salt Works, 40 km south of Dandi. However, Gandhi was arrested on the midnight of 4–5 May 1930, just days before the planned action at Dharasana. The Dandi March and the ensuing Dharasana Satyagraha drew worldwide attention to the Indian independence movement through extensive newspaper and newsreel coverage. The satyagraha against the salt tax continued for almost a year, ending with Gandhi's release from jail and negotiations with Viceroy Lord Irwin at the Second Round Table Conference.[3] Although over 60,000 Indians were jailed as a result of the Salt Satyagraha, the British did not make immediate major concessions.

1942 – Quit India Movement

The Quit India Movement (translated into several Indian languages as the Leave India Movement), also known as the August Movement, was a movement launched at the Bombay session of the All-India Congress Committee by Mahatma Gandhi on 9 August 1942, during World War II, demanding an end to British rule in India.

Pic 4: Quit India Movement

After the failure of the Cripps Mission to secure Indian support for the British war effort, Gandhi made a call to Do or Die in his Quit India speech delivered in Bombay on 8 August 1942 at the Gowalia Tank Maidan. The All-India Congress Committee launched a mass protest demanding what Gandhi called "An Orderly British Withdrawal" from India. Even though it was at war, the British were prepared to act. Almost the entire leadership of the Indian National Congress was imprisoned without trial within hours of Gandhi's speech. Most spent the rest of the war in prison and out of contact with the masses. The British had the support of the Viceroy's Council (which had a majority of Indians), of the All India Muslim League, the Hindu Mahasabha, the princely states, the Indian Imperial Police, the British Indian Army, and the Indian Civil Service. Many Indian businessmen profiting from heavy wartime spending did not support the Quit India Movement. Many students paid more attention to Subhas Chandra Bose, who was in exile and supporting the Axis Powers. The only outside support came from the Americans, as President Franklin D. Roosevelt pressured Prime Minister Winston Churchill to give in to some of the Indian demands. The

Quit India campaign was effectively crushed. The British refused to grant immediate independence, saying it could happen only after the war had ended.

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B.2: Observations and Learning's:

Understood usage of different text styles in Libre Office.

B.3: Conclusion:

In conclusion, introduction to Libre Office tools was successfully comprehended.